EFU TEXTILE LICENCE v1.0

Experimental EFU-based Textile Industry & Sustainability Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for sustainable textile manufacturing (apparel, home textiles), supply chains, and recycling. **NOT** a legal requirement, EU EPR (Extended Producer Responsibility) substitute, or certification (GOTS, Oeko-Tex) replacement, but an open research protocol measuring textile resource use in human-scale EFU metrics (water, materials, energy, waste).

I. Core Concepts

1. **EFU-Textile (EFU-TEX)**
   EFU framework application for textile industry, comprising:
   - **EFU-W_TEX**: water use (dyeing, pre-treatment, washing)
   - **EFU-M_TEX**: material consumption (fibers, yarns, finished products)
   - **EFU-H_TEX**: energy use (spinning, weaving, sewing, drying)
   - **EFU-S_TEX**: entropy/microplastic waste
   - **EFU-C_TEX**: carbon footprint (full supply chain emissions)
2. **Textile Sovereignty Gap (SS_TEX)**
   SS_TEX = TEFU_TEX_total – Regional Textile Carrying Capacity (RTCC_TEX)
   Measures if textile manufacturing stays within regional water/biocapacity regeneration limits.
3. **MROI_TEX**
   Textile Metabolic Return on Investment = THI_TEX / TEFU_TEX
   THI_TEX = clothing coverage + local jobs + recycling rate + design innovation.

II. EFU Textile Licence – Research Status

The licence documents EFU-audited textile manufacturing chains with standardized, public metrics.
Does NOT replace EU textile EPR, Digital Product Passport, or national waste regulations.
Enables textile factories/brands benchmarking in EFU research context.

III. Metrics and Thresholds (Textile-specific – experimental)

1. EFU Components (per kg garment or per piece)

EFU-W_TEX (Water)
EFU-W_TEX = water_use_m3 / 18.25 / garment_kg
Example: Cotton t-shirt = 2,700 liters → 2700/18.25/0.2kg = 0.74 human-years/kg

EFU-M_TEX (Materials)
EFU-M_TEX = (fiber_kg + dye_kg + auxiliaries) / 30.0 / garment_kg
Baseline: 30 kg/person/year textile consumption

EFU-H_TEX (Energy)
EFU-H_TEX = manufacturing_transport_energy_MWh / 12.88 / garment_kg

EFU-S_TEX (Entropy/Microplastics)
EFU-S_TEX = washing_microplastic_equivalent / 11500 / garment_kg
Experimental: wash cycles × fiber shedding factor

TEFU_TEX = 1.5×EFU-W + 1.2×EFU-M + 1.0×EFU-H + 0.8×EFU-S + 0.5×EFU-C

2. Regional Textile Carrying Capacity (RTCC_TEX)

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RTCC_TEX = (Sustainable_water_m3_year / Population) / 18.25 m3/person/year baseline
Example: Bangladesh = 100B m3 / 170M / 18.25 = 0.032 EFU_TEX/person/year
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IV. Licence Categories (EFU-TEX Labels)

Level	SS_TEX Condition	MROI_TEX	Label	Interpretation
1	SS_TEX ≤ 0	≥1e-3	**EFU-TEX SOVEREIGN**	Within water/biocapacity
2	0 < SS_TEX ≤ 10k EFU	≥1e-4	**EFU-TEX BALANCED**	Monitorable
3	10k < SS_TEX ≤ 50k EFU	≥1e-5	**EFU-TEX DEFICIT**	Significant overload
4	SS_TEX > 50k EFU	<1e-5	**EFU-TEX CRITICAL**	Structurally imbalanced

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V. Regional RTCC_TEX Calibration Examples

Country/Region	Annual Water (B m³)	Population (M)	RTCC_TEX (EFU/person/year)
BANGLADESH	100	170	0.032
CHINA	2,800	1,410	0.109
INDIA	1,900	1,430	0.073
VIETNAM	300	100	0.164
EU AVERAGE	300	448	0.037

Calibration: `RTCC_TEX = Water_Bm3 / Population_M / 18.25 m3/person/year`

VI. Example: Bangladesh Cotton T-Shirt Factory

Input (annual factory):

- Production: 1M t-shirts (200g each = 200 tonnes)
- Water: 500M liters (500,000 m³)
- Energy: 25,000 MWh
- Workers: 5,000 staff
- Region: Bangladesh (RTCC_TEX = 0.032 EFU/person/year)

Calculation:

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EFU-W_TEX = 500,000/18.25/200,000 = 0.137 human-years/kg
EFU-H_TEX = 25,000/12.88/200,000 = 0.010 human-years/kg
EFU-M_TEX = 240,000 kg cotton/30/200,000 = 0.040 human-years/kg

TEFU_TEX_total ≈ 37,400 EFU/year
Allocated = 5,000 × 0.032 = 160 EFU/year
SS_TEX = 37,400 – 160 = +37,240 EFU/year (Level 4: CRITICAL)

Result: EFU-TEX CRITICAL
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VII. Implementation Notes

1. **Branding**: EFU-TEX labels only after documented EFU audit (104.3 compliance)
2. **EU EPR complement**: TEFU_TEX metric foundation for eco-modulation (fees based on water/material use)
3. **Data sources**: Tier 1 = water meters, energy bills; Tier 2 = manufacturer specifications